

The Cross-Section of Factor Returns

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Abstract

We explore the cross-section of factor returns using a sample of 150+ equity factors. Most factors exhibit a positive premium and a negative market beta in the long run. Factor themes with a clear positive beta, in particular low leverage and size, have no alpha after controlling for this beta exposure. The remaining factors generate most of their raw return in bear markets, which also explains half of their decay in the predominantly bullish post-2004 period. Beta-adjusting factor returns yields alphas that are not only higher but also considerably more stable. We also revisit factor performance cyclicity, establish a low-beta effect at the level of factors, and confirm the existence of seasonal and momentum effects in the cross-section of factor returns. Altogether, our insights into factor behavior aid the development of more robust factor-based investment strategies.

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1. Introduction

Since the discovery of the size, value, and momentum effects in the 1980s and 1990s, more and more priced factors have been identified in the asset pricing literature. In recent years the focus has shifted from establishing yet another factor to reflecting on the entire ‘zoo’ of factors that has emerged by now. This paper adds to this new stream of literature by deepening the general understanding of the cross-section of equity factor returns. Whereas the classic Fama and French (1993, 2015) asset pricing factors, such as size (SMB), value (HML), and momentum (WML), are well understood after decades of extensive research, much less is known about the many other factors in the zoo. We are particularly interested in the long-term performance of factors, their risk characteristics and behavior, the cyclicalities of factor returns, and the potential of factor rotation strategies. The insights obtained can aid the design of more robust factor-based investment strategies, help shape the agenda for future factor research, and may also have implications for factor portfolio construction methodologies.

Our work is related to recent studies that have examined questions such as how many factors are really needed, the replicability of originally reported results, and the decay of factor performance over time. For instance, Bessembinder, Burt, and Hrdlicka (2022) find that the variation in the returns of hundreds of factors can be captured with a much smaller number of principal components, implying that the zoo can be substantially reduced, although not to just a handful of factors as in common asset pricing models. Other influential critical studies include Hou, Xue, and Zhang (2020), who find that many anomalies cannot be replicated when adhering to the standard factor construction methodology that mitigates the impact of illiquid micro-cap stocks, and McLean and Pontiff (2016), who report a 58% lower performance after publication for US factors. On the other hand, Jacobs and Müller (2020) find little evidence of factor performance decay for international equity markets, and Jensen, Kelly, and Pedersen (2021) challenge the notion that there is a replication crisis in finance. While these important debates remain unsettled, many other interesting things are going on in the cross-section of factor returns.

Our main insights can be summarized as follows. Most factors exhibit a positive premium, although some do not pass this basic test. Most factors also exhibit a negative long-term market beta, while the only two themes with a clear positive beta, namely low leverage and size, have no alpha after controlling for this beta exposure. The negative beta of the remaining factors has major implications for their raw return in different market environments, with most of the performance being generated in bear markets. This suggests that mispricing typically builds up during bull markets (resulting in weak factor returns) and tends to get corrected in bear markets (resulting in large factor payoffs). This result also sheds a new light on factor performance decay, as we estimate that about half of the decay after 2004 can be attributed to a decline in bear markets during

this period. Factor alphas, i.e. their returns adjusted for the ex post beta exposure, are not only more stable but also higher. Our study also deepens the insights into factor performance cyclicalities, establishes a low-beta effect at the level of factors, and confirms the existence of seasonal and momentum effects in the cross-section of factor returns. Altogether, our findings show which kind of factors are rewarded and which ones not, how their performance varies in different market environments, and how investment strategies can be built on factors themselves.

The remainder of this paper is organized as follows. We first discuss the data and methodological set-up. After that we subsequently examine the long-term performance of factors, factor performance in different economic environments, and various factor rotation strategies. Finally, we conclude.

2. Data

Various broad factor libraries have been made available online, such as the Open Source Asset Pricing library of Chen and Zimmermann (2020), the Q factor library of Hou, Xue, and Zhang (2020), and the Global Factor Data library of Jensen, Kelly, and Pedersen (2021).¹ A drawback of Open Source Asset Pricing is that a holding period of 12 months is imposed for all the series that follow the standard 2x3 sorting methodology of Fama and French (1993), which vastly underestimates the premiums on short-term factors such as 1-month return reversal. The Q factor library has a bit shorter history and uses non-standard 3x5 sorts. Moreover, both databases only cover the U.S. equity market.

Based on these considerations we deem the Jensen, Kelly, and Pedersen (2021) factor library to be most suitable for the purposes of this paper. We examine all the factors in their Global Factor Data library, which gives a total number of 153 factors, classified in 13 themes using statistical clustering techniques. We focus on the U.S. market because it offers the longest history. We obtain monthly top-minus-bottom portfolio return series constructed with the ‘capped value weighted’ methodology recommended by the authors. Value weighting prevents an excessive weight of micro-cap stocks, similar to the Fama-French 2x3 methodology, while the capping additionally ensures that the results are not dominated by a handful of mega-cap stocks, without arbitrarily inflating the weight of small-cap stocks to 50% as with the Fama-French approach. The sample period is from July 1963 to December 2021, which gives coverage for 90% of the factors at the start, and full coverage from 1971 onwards.

¹ See <https://www.openassetpricing.com/>; <http://global-q.org/index.html>; <https://jkpfactors.com/>.

3. Long-term factor performance

We first examine the full-sample raw return of the 153 factors in our dataset. Exhibit 1 shows that most factors offer a positive premium, which is not really surprising because this is a defining characteristic of priced factors. Nevertheless, 11% of the factors in the sample display a negative average return. This might be due to various reasons, such as post-publication decay or differences in the factor construction methodology. For instance, Hou, Xue, and Zhang (2020) find that many anomalies cannot be replicated when mitigating the impact of illiquid microcap stocks. The unrewarded factors can be found across all themes, but the low-leverage theme appears to be particularly weak, with a negative average return for 5 out of the 11 factors, and an average premium of just 0.4% per annum.

INSERT EXHIBIT 1 ABOUT HERE

However, we should not just look at raw returns to decide whether to accept or dismiss factors. For instance, various low-risk factors, such as past 60-month beta, may appear weak based on their raw return, but these factors are obviously being short-changed by ignoring that a zero return difference between low-beta and high-beta stocks does, in fact, indicate a major market inefficiency. On the other hand, a positive raw factor return is not necessarily evidence of an anomaly, if the factor simply has a tilt towards stocks with higher betas that ought to earn higher returns, at least according to the Capital Asset Pricing Model (CAPM). In order to shed light on this Exhibit 2 reports the full-sample beta of each factor.

INSERT EXHIBIT 2 ABOUT HERE

Interestingly, the vast majority of factors (75%) is associated with a negative market beta. For the low-risk factors this is by design and therefore most pronounced, but also many factors in the value, investment, profitability, and momentum themes display negative betas. Only two themes show predominantly positive betas, namely low leverage and size. Exhibit 3 extends this analysis by reporting 1-month lagged beta exposures, which follow from regressing factor returns not only on contemporaneous but also on 1-month lagged market returns in excess of the risk-free return. Although these lagged beta exposures are also mostly negative (69%), their magnitude is generally small. The main exception are the size factors, which do show non-negligible positive lagged beta exposures, consistent with the findings of Asness et al. (2018) and Blitz and Hanauer (2021).

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Exhibit 4 shows the annualized CAPM alphas of the factors, where for simplicity we only adjust for the contemporaneous market exposure. We observe a clear dichotomy between

the first eleven themes and the final two, low-leverage and size. Whereas 95% of the alphas are positive among the former, the alpha is clearly negative on average for the latter. Interestingly, these are the same two themes that were characterized by positive betas. The absence of positive alphas for the low leverage and size factors implies that they do not qualify as priced factors. For the size factor this conclusion is in line with Blitz and Hanauer (2021), who also reject the existence of a size premium. In light of these results we consider the first eleven themes to be most relevant, although we will continue to report results for low leverage and size for completeness.

INSERT EXHIBIT 4 ABOUT HERE

Looking at the individual factors within all thirteen themes we observe some outliers. For example, R&D/ME is included in the size theme, but shows a decent alpha of 3.3% per annum, roughly at par with proper factors. This may indicate a misclassification. Since R&D/ME measures investment in knowledge capital, it might be better to treat it as a quality or growth factor. Another notable example is Debt/ME, which is classified as a value factor, but underperforms the other value factors, barely showing any alpha at all. Although value factors are generally defined as some kind of fundamental value metric divided by market value, Debt/ME is probably better understood as a leverage or risk factor. Thus, we see some room for improvement in how factors are assigned to themes. But since this is not vital for the purposes of this paper we will stick to the default classification in the remainder.

Exhibit 5 plots the alphas of all the factors against their betas, which reveals a strong inverse relation. Thus, low-beta factors generally have the highest alphas, while high-beta factors generally display the lowest alphas, or actually no alpha at all. A regression shows that this relation is highly statistically significant (t-statistics between brackets):

$$\alpha = 2.50\% - 7.70\% \beta + \varepsilon \quad (1)$$

(13.3) (-10.3) R-squared: 41.1%

The interpretation of this result is that for every 0.01 change in the beta, the annualized alpha of a factor changes by 7.7 basis points in the opposite direction. For factors with betas above 0.32 the expected alpha even becomes negative.

INSERT EXHIBIT 5 ABOUT HERE

Exhibit 6 shows another, even stronger inverse relation. The graph plots equity betas against bond betas, estimated by regressing factor returns simultaneously on equity market excess returns and the excess returns of 10-year U.S. Treasury bonds. Equity betas and bond betas turn out to be almost mirror images of each other, i.e. factors with low (negative) equity betas tend to have high (positive) bond betas, and vice versa. Thus, factors with a negative equity beta tend to benefit from falling interest rates, while factors

with a positive equity beta tend to benefit from rising interest rates. This confirms the conclusions of Blitz (2020), who found a similar result for a limited set of factor and industry portfolios. The correlation between the two betas is an astounding -90%, and also when we formally regress the bond betas of factors on their equity betas the result is extremely significant (t-statistics again between brackets):

$$\beta_{\text{bond}} = -0.03 \quad -0.62 \beta_{\text{equity}} + \varepsilon \quad (2)$$

(-4.9) (-25.5) R-squared: 81.1%

INSERT EXHIBIT 6 ABOUT HERE

We next examine which factors exhibit a January effect. To this end Exhibit 7 reports average annualized factor alphas in January months only. On average, across all factors, January does not look like a particularly eventful month, but at the level of the individual factors and themes we do observe quite irregular behavior in January months. Most notably the size factors, which have average January alphas in the range of 20-30% per annum. This finding is in line with the existing literature, e.g. Alquist, Israel, and Moskowitz (2019). Value and investment factors show above-average alphas, while low-risk, profitability, and momentum factors generally show negative alphas. Altogether, this characterizes January months as mini junk rallies, with strong performance of small, cheap, risky, unprofitable, beaten down stocks.

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4. Performance in different economic environments

The negative equity betas of most factors, in particular the ones that actually provide alpha, suggest that raw factor returns may be higher on average in bear markets than in bull markets. In order to test this conjecture, we examine the average raw factor returns in these two different environments, using the bull versus bear market classification of Geertsema and Lu (2022), who follow the methodology of Pagan and Sossounov (2003). This classification method first identifies local peaks and troughs, and then applies censoring rules such as a minimum required length of a bull/bear cycle.

Exhibit 8 confirms our intuition as the average annual return across all factors is a mere 0.9% in bull markets, versus a solid 8.6% in bear markets. In other words, factors appear to generate most of their return in bear markets, while barely delivering in bull markets. Since bear markets comprise only 21% of the sample, this indicates that factor payoffs tend to be quite skewed. The results suggest that mispricing builds up gradually during prolonged bull markets (resulting in weak factor returns) and tends to get corrected relatively quickly in bear markets (resulting in large factor payoffs). This strong

asymmetry highlights the importance of making beta adjustments. Indeed, Exhibit 9 shows that the alpha of factors in bull versus bear markets is much more balanced.

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The pronounced market beta exposures of many factors, and the major implications for performance in different market environments, questions the methodology used to construct factor portfolios. Clearly, having long weights and short weights that both add up to 100%, i.e. dollar neutrality, does not suffice for establishing true market neutrality. Only for low-risk factors it is common to address this issue, in particular by leveraging the low-risk long portfolio and deleveraging the high-risk short portfolio to betas of 1 each, as with the BAB (betting against beta) factor of Frazzini and Pedersen (2014) and the VOL factor of Blitz, van Vliet, and Baltussen (2020). For other factors great care is taken to ensure size neutrality, but our results indicate that improved market neutrality may be at least as important. However, it remains to be seen whether better ex ante beta neutrality of factors also translates into significantly better ex post beta neutrality, and whether it is effective at turning the alphas that we now observe ex post into higher raw returns.

Our insights also shed a fresh perspective on the widely documented decay of factor performance over time. In line with studies such as McLean and Pontiff (2016), we find that the average raw factor return is substantially lower after 2004 compared to before 2004. However, the market was in a bear market state 27% of the time before 2004, compared to only 9% of the time after 2004, which amounts to a factor 3 difference. The total change in factor performance after 2004 can be broken down into the change in factor performance during bull markets, the change in factor performance during bear markets, and the impact from the change in the bull versus bear market frequency. Exhibit 10 shows that although factor performance decayed in bull market as well as bear market environments, about half of the decay in overall performance can actually be attributed to the much lower frequency of bear markets in the post-2004 period. Adjusted for this change in market environments, the average performance decay is only 33% instead of 67%. Thus, about half of the decay in recent factor performance may be attributed to the market being in a predominantly bullish state, which tends to be less favorable for raw factor returns.

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The binary distinction between bull and bear markets is rather crude. Blitz (2022) defines a Quant Cycle and finds that factors do well in bear markets caused by the bursting of growth bubbles, but not in bear markets caused by a crash of value stocks. And although factors struggle in extreme bull markets driven by rallies of growth stocks (as in 1998-

1999) or junk stocks (like tech in 2003 and financials in 2009), they actually do fine during moderate bull markets. The Quant Cycle is inspired by the finding that traditional business cycle indicators fail to capture the large variation in factor returns, and that major turning points of factors seem to be caused instead by abrupt changes in investor sentiment that do not appear to be linked to the macroeconomy. The cycle is inferred directly from factor returns, and consists of a normal stage that is occasionally interrupted by drawdowns of the value factor that are in turn followed by reversals. Value factor drawdowns can occur in bullish environments due to growth rallies and in bearish environments due to crashes of value stocks. Also for the reversals a distinction can be made between a bullish subvariant (junk rallies) and a bearish one (growth crashes).

Exhibit 11 shows the average annualized alpha of all 153 factors in our sample across the different stages of the Quant Cycle. Consistent with the findings in Blitz (2022), who only considers 8 factors, average factor returns are solid in the normal stage, well above average during bear reversals, but weak in growth rallies, value crashes, and bull reversals. The main difference is that Blitz (2022) found a strong negative average factor performance in bull reversals, which are characterized by crashes of the momentum factor, whereas the impact of those is subdued in our sample where momentum is just one of thirteen themes, resulting in a roughly flat average factor performance instead. Importantly, factors do well in the normal stage which prevails about two-thirds of the time and is characterized by moderately rising markets, implying that the binary bull versus bear markets split considered before is too simplistic.

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Exhibit 12 compares factor performance during growth rallies and value crashes with factor performance during bull reversals. The graph shows a strong inverse relation (correlation -62%) between factor performance in these two environments, so factors that do well during growth rallies/value crashes tend to do poorly during bull reversals, and vice versa. Low-risk factors are an exception, albeit not in a positive sense, as their returns are negative in both phases. Although this might suggest that low-risk factors are most problematic, they actually provide the highest average alpha during the normal stage that is most common. The graph also shows that few factors manage to do well during both growth rallies/value crashes and bull reversals. These are mostly factors from the low leverage and size themes, that we previously dismissed because they fail to generate alpha in the long run. Altogether, these results imply that it is challenging to find all-weather factors that do well in all phases of the cycle.

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Another indicator of factor sensitivity to the Quant Cycle is the volatility of the mean alphas in the different stages. For a factor with the exact same average alpha in each stage

of the cycle this volatility would be zero, while for a factor that does very well in some stages but very poorly in others it will be very high. Exhibit 13 shows that these volatilities display a U-shaped pattern. Value-like factors on the left side of the graph and momentum-like factors on the right side of the graph have a high cycle volatility, but factors in the middle such as short-term reversal and seasonals appear to be largely immune to the turbulence of the Quant Cycle. The drawback of these factors is that they are relatively fast and thus involve a high turnover. This concern is addressed by Blitz et al. (2022), who find that a significant net return can be obtained by integrating multiple short-term factors and applying cost-mitigating buy-sell rules. Moreover, this short-term alpha is largely orthogonal to the traditional Fama-French asset pricing factors.

INSERT EXHIBIT 13 ABOUT HERE

5. Factor rotation strategies

In this section we consider several factor rotation strategies, namely a low-beta approach, a seasonal trading rule, and factor momentum. The low-beta strategy sorts factors into quintile portfolios every month, based on the market beta exposure of their raw returns estimated over the prior 60 months. Exhibit 14 shows the performance of these quintile portfolios of factors over the complete sample period. We observe that the ex post betas of the quintile portfolios vary from about -0.5 to 0.2. Not surprisingly, the lowest beta portfolio contains many factors from the low-risk theme, and, consistent with our earlier findings, the low leverage and size factors end up mostly in the highest beta portfolio. The long-term relation between beta and average raw return turns out to be flat, but adjusting for the ex post market beta exposure gives a strongly inverse relation between alpha and beta.

INSERT EXHIBIT 14 ABOUT HERE

In other words, the factors with the lowest betas tend to offer the highest alphas, consistent with what we saw before when examining the relation between full-sample (ex post) betas and alphas. This is further illustrated in Exhibit 15, which shows the cumulative alpha of the beta-sorted factor quintile portfolios over time, and can be interpreted as yet another manifestation of the well-known low-risk anomaly (see e.g. Blitz, van Vliet, and Baltussen, 2020). However, the alpha spread is similar in magnitude and highly correlated with the straightforward low-risk factors in our sample, which indicates that the strategy is effectively capturing the same alpha in a different way.

INSERT EXHIBIT 15 ABOUT HERE

The seasonal strategy selects factors based on their historical returns in the same calendar month using a lookback period of 20 years. Thus, in January we go long the factors with

the highest return in the past 20 Januaries and short the factors with the lowest return in the past 20 Januaries, and similarly for the other calendar months. This approach follows Keloharju, Linnainmaa, and Nyberg (2016), who apply it to individual stocks, industries, and a small number of factor-sorted portfolios. Exhibit 16 shows the performance of the equally-weighted top-minus-bottom quintile portfolio of factors. The seasonal strategy has a long-term annualized return of 4.7% with a Sharpe ratio of 0.43. However, the graph shows that most of this performance is realized in the first half of the sample, while performance over the second half of the sample is much flatter. Thus, while the seasonal factor rotation strategy was clearly effective in the past, it is questionable whether it is still viable.

INSERT EXHIBIT 16 ABOUT HERE

We next consider a factor momentum strategy, where every month we go long the factors with the highest return over the preceding month, and short the factors with the lowest return over the preceding month. As shown by Gupta and Kelly (2019), the profitability of such a strategy stems from the significant autocorrelation in monthly factor returns. Exhibit 17 reports the 1-month autocorrelations of the raw factor returns, and we observe that these are indeed predominantly positive, with an average across all factors of 0.09 and a positive sign in 88% of the cases. Only two factor themes do not appear to exhibit positive autocorrelation, namely momentum and short-term reversal. Interestingly, the factors in these two themes have in common that they are entirely based on past stock returns, unlike most of the other factors which also make use of financial statement data to a bigger or smaller extent.

INSERT EXHIBIT 17 ABOUT HERE

Exhibit 18 shows the return of the equally-weighted top-minus-bottom quintile portfolio of factors based on past 1-month factor returns. In line with Gupta and Kelly (2019) and Arnott et al. (2020) we find a strong performance for the factor momentum strategy, with an annualized return of 10% and an associated Sharpe ratio of 0.58. These numbers hardly change when excluding the momentum and short-term reversal factors, so it does not seem to matter much if the factors without autocorrelation in their returns are dropped from the analysis. Contrary to the seasonal factor strategy, the factor momentum strategy does not show a strong performance decay over time. Thus, factor momentum has remained a profitable strategy.

INSERT EXHIBIT 18 ABOUT HERE

The performance of the three factor rotation strategies in the different stages of the Quant Cycle is shown in Exhibit 19. The low-beta strategy shows a strong positive alpha during bear reversals but large negative alphas during growth rallies, value crashes, and bull reversals, which is very similar to the pattern observed before for factors in the low-risk

theme. However, the seasonal and factor momentum strategies are able to navigate the Quant Cycle with much less difficulty. The 1-month factor momentum strategy appears to be particularly robust, with a solid return in all stages. It even manages to do well in both growth rallies (26% annualized return) and bull reversals (35% annualized return), unlike any of the 153 individual factors in the zoo. This provides another illustration that short-term signals can be the key to escaping the huge cyclicity of traditional factors.

INSERT EXHIBIT 19 ABOUT HERE

6. Conclusion

We study the cross-section of equity factor returns using the data library of Jensen, Kelly, and Pedersen (2021). Most of the 150+ factors in the sample exhibit a positive premium and a negative long-term market beta. The only two themes with a clear positive beta, namely low leverage and size, have no alpha after controlling for this beta exposure. The remaining factors generate most of their raw return in bear markets, and half of their decay post-2004 can be attributed to bear markets being considerably less prevalent in this period. Beta-adjusted returns, i.e. alphas, are considerably more stable, indicating that factor construction methodologies may be improved by going beyond mere dollar and size neutrality. Bond betas are very strongly inversely related to equity betas. Our study also deepens the Quant Cycle results of Blitz (2022), establishes a low-beta effect at the level of factors, and confirms the existence of seasonal and momentum effects in the cross-section of factor returns. Exhibit 20 gives an overview of some of the main findings at the level of factor themes. Altogether, our insights deepen the understanding of factor behavior and can aid the development of more robust factor-based investment strategies.

INSERT EXHIBIT 20 ABOUT HERE

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Exhibit 1: Annualized raw factor returns, 1963:07-2021:12

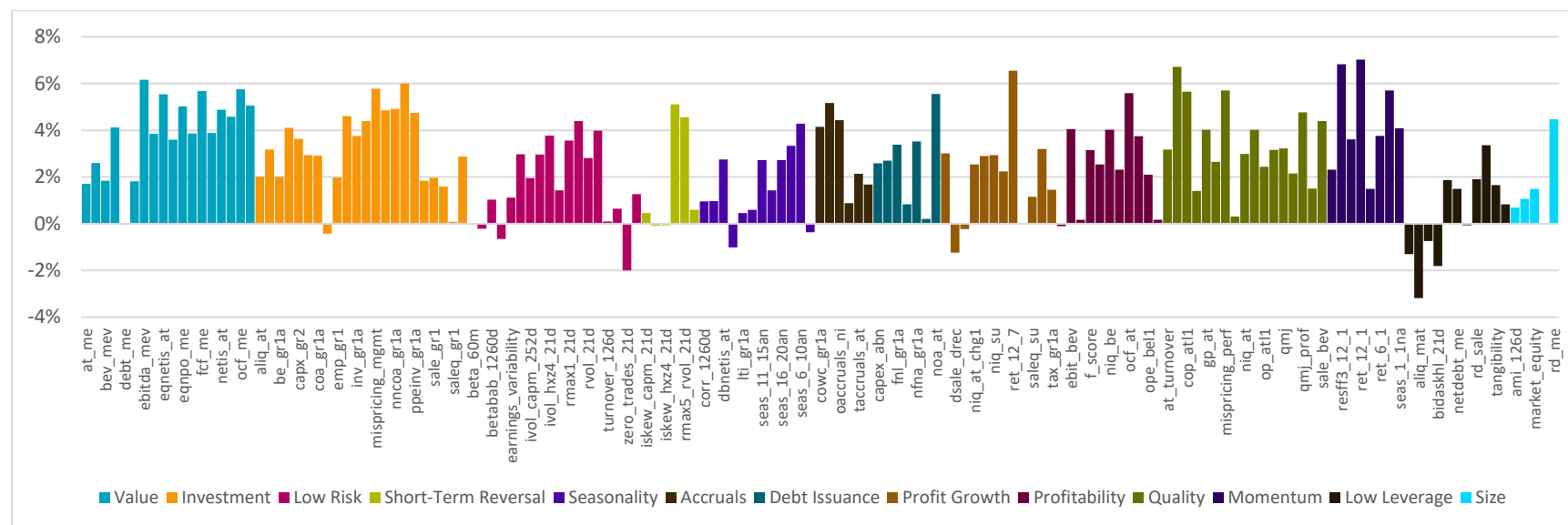


Exhibit 2: Market beta exposure of factors, 1963:07-2021:12

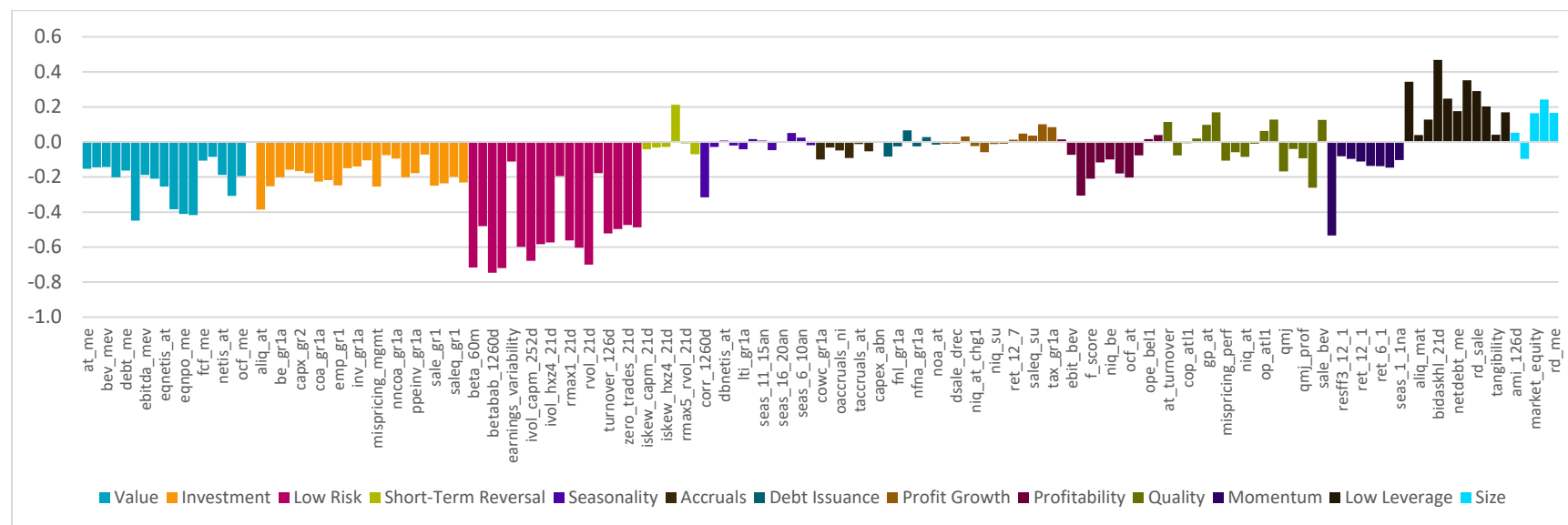


Exhibit 3: Lagged market beta exposure of factors, 1963:07-2021:12

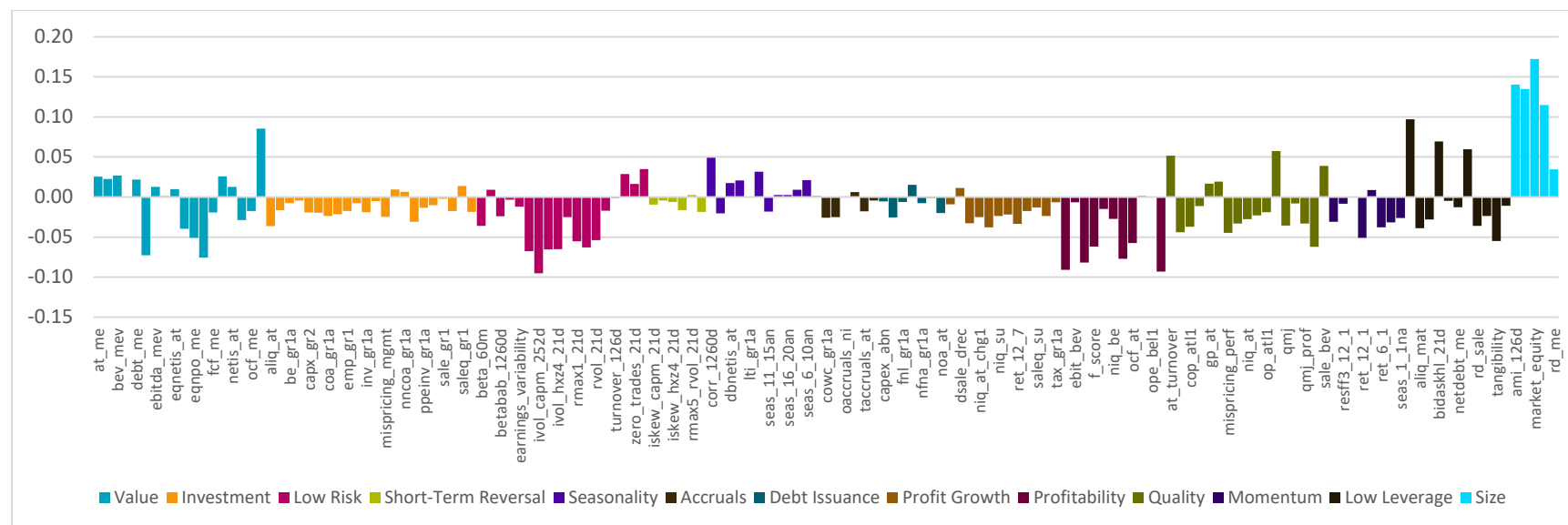


Exhibit 4: Annualized factor alphas, 1963:07-2021:12

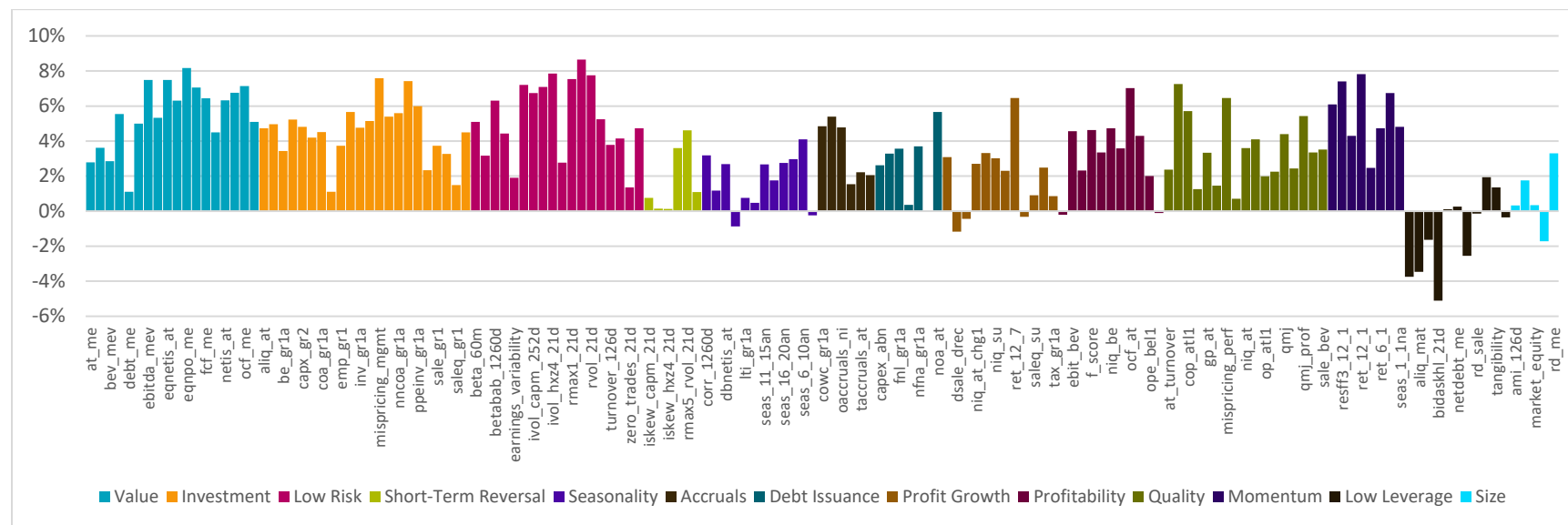


Exhibit 5: Alphas versus betas of factors, 1963:07-2021:12

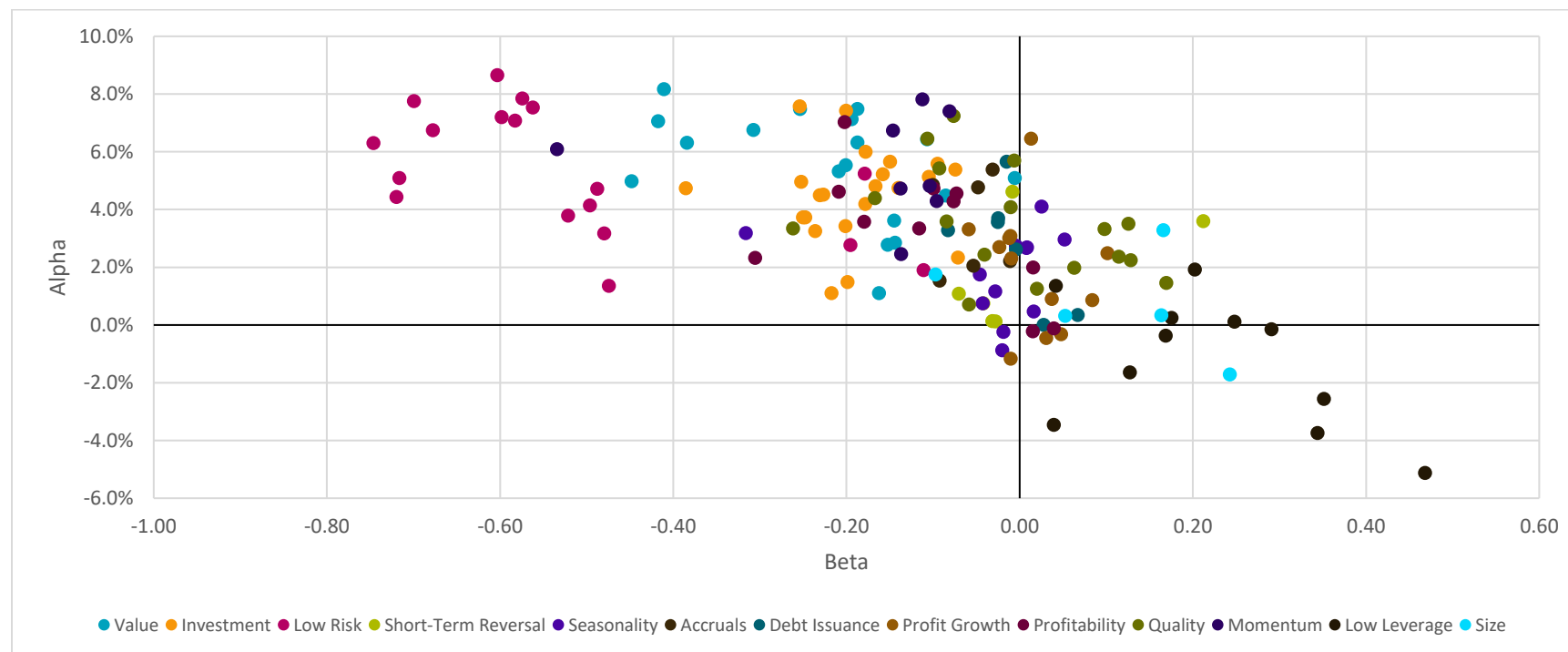


Exhibit 6: Equity beta versus bond beta of factors, 1963:07-2021:12

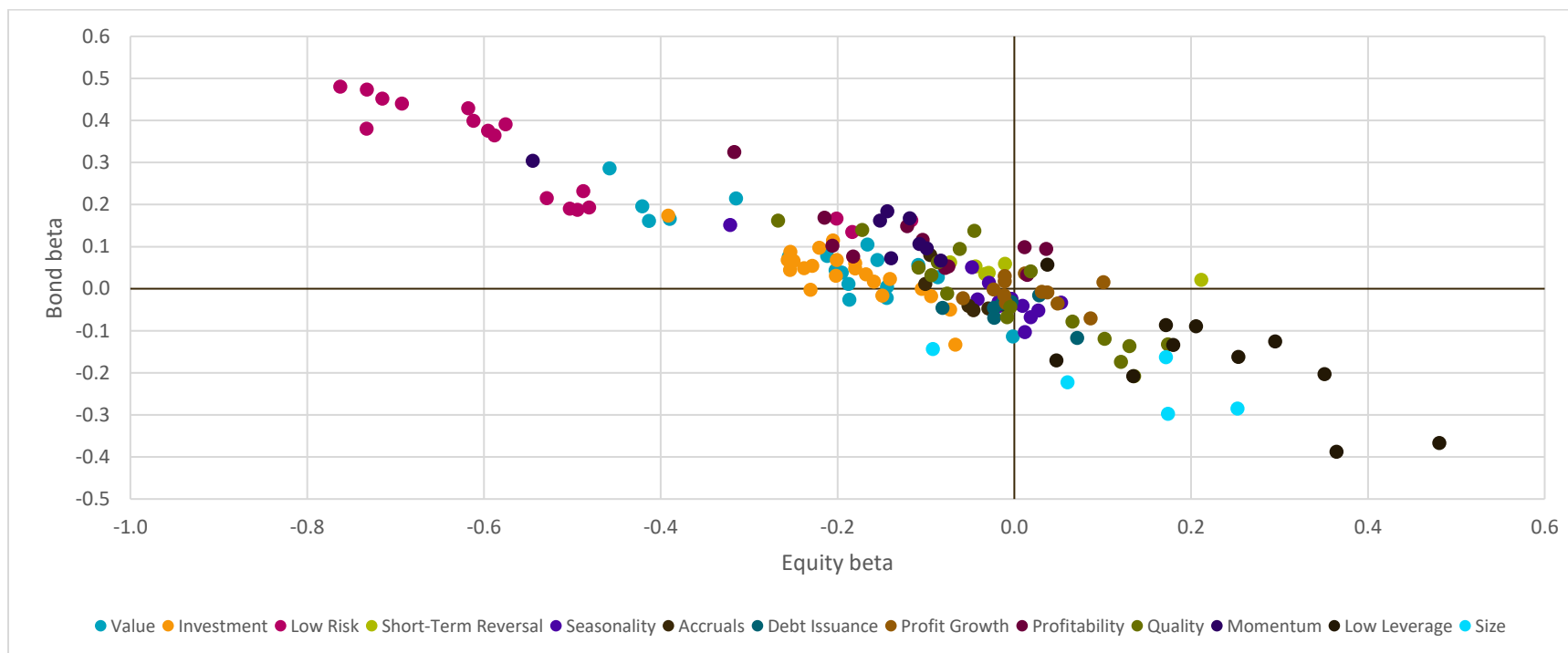


Exhibit 7: Annualized factor alphas in January months only, 1963:07-2021:12

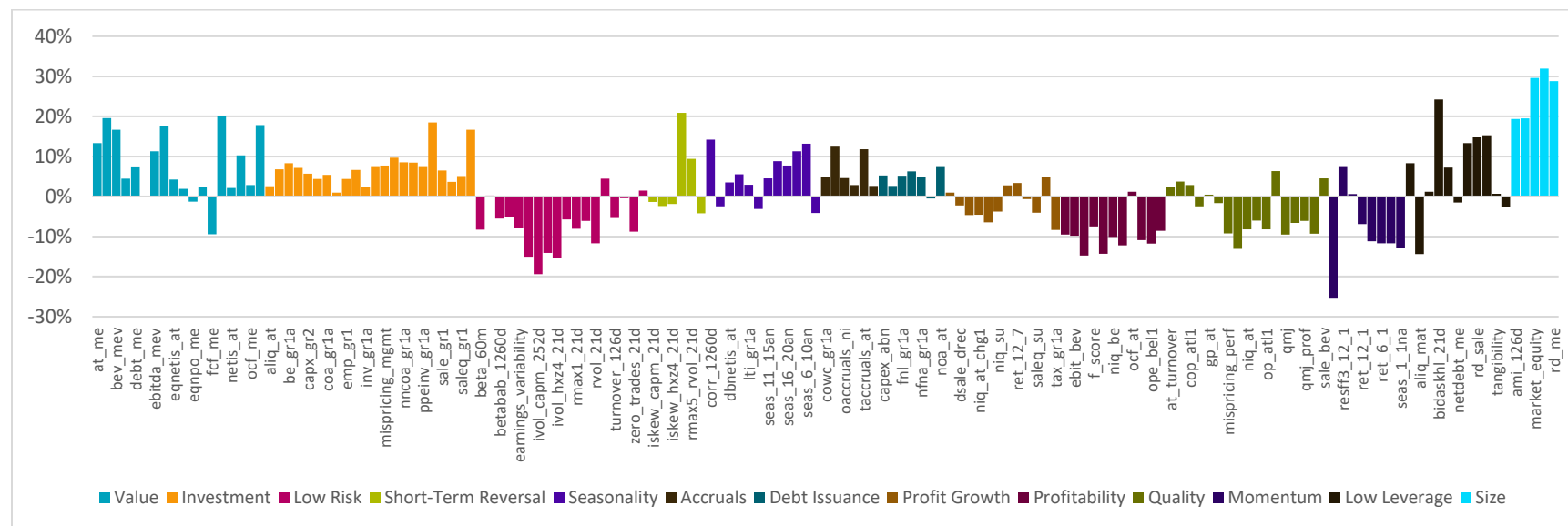


Exhibit 8: Annualized raw return of factor themes in bull versus bear market periods, 1963:07-2021:12

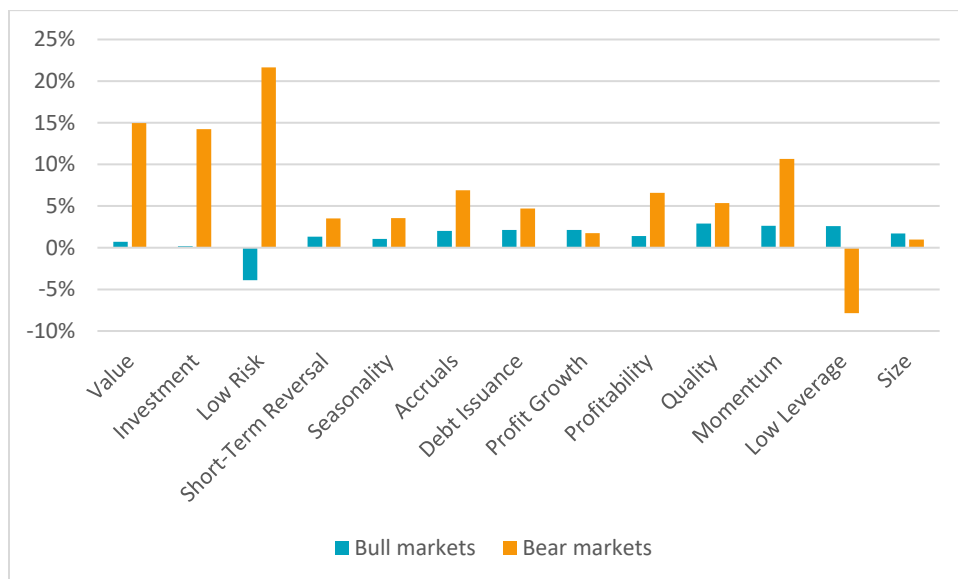


Exhibit 9: Annualized alpha of factor themes in bull versus bear market periods, 1963:07-2021:12

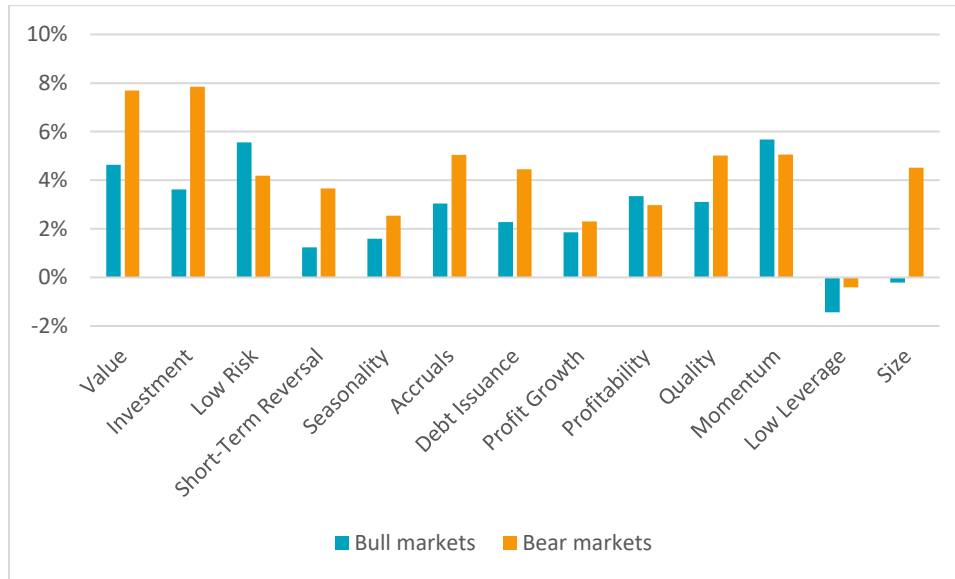


Exhibit 10: Breakdown of factor performance decay, 1963:07-2021:12

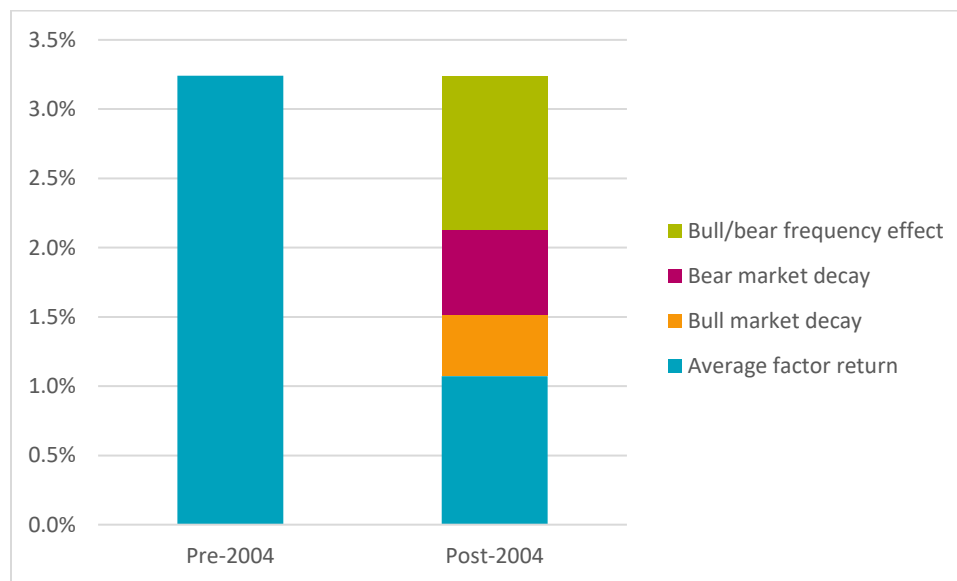


Exhibit 11: Average factor alpha over the Quant Cycle, 1963:07-2021:12

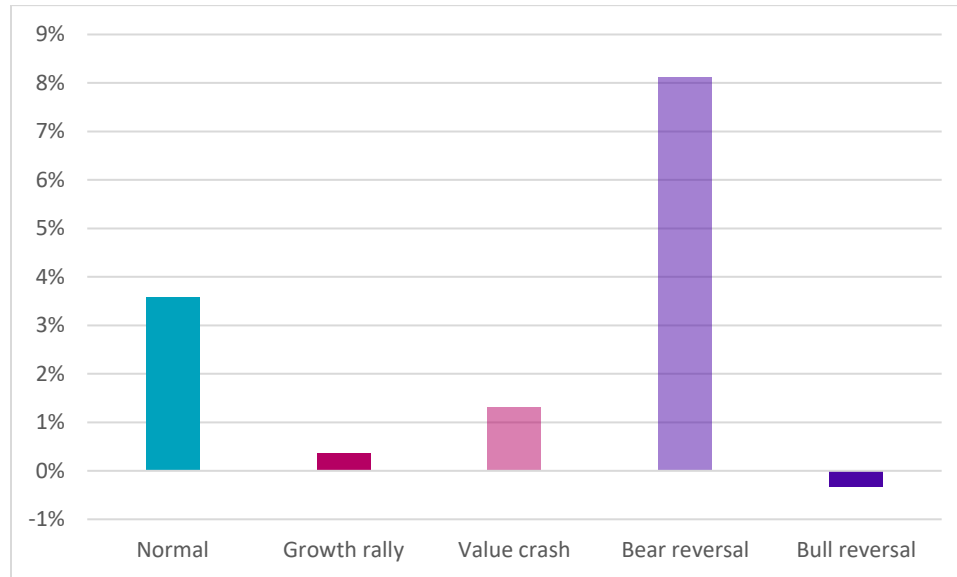


Exhibit 12: Comparison of factor alphas in growth rallies (or value crashes) versus bull reversals, 1963:07-2021:12

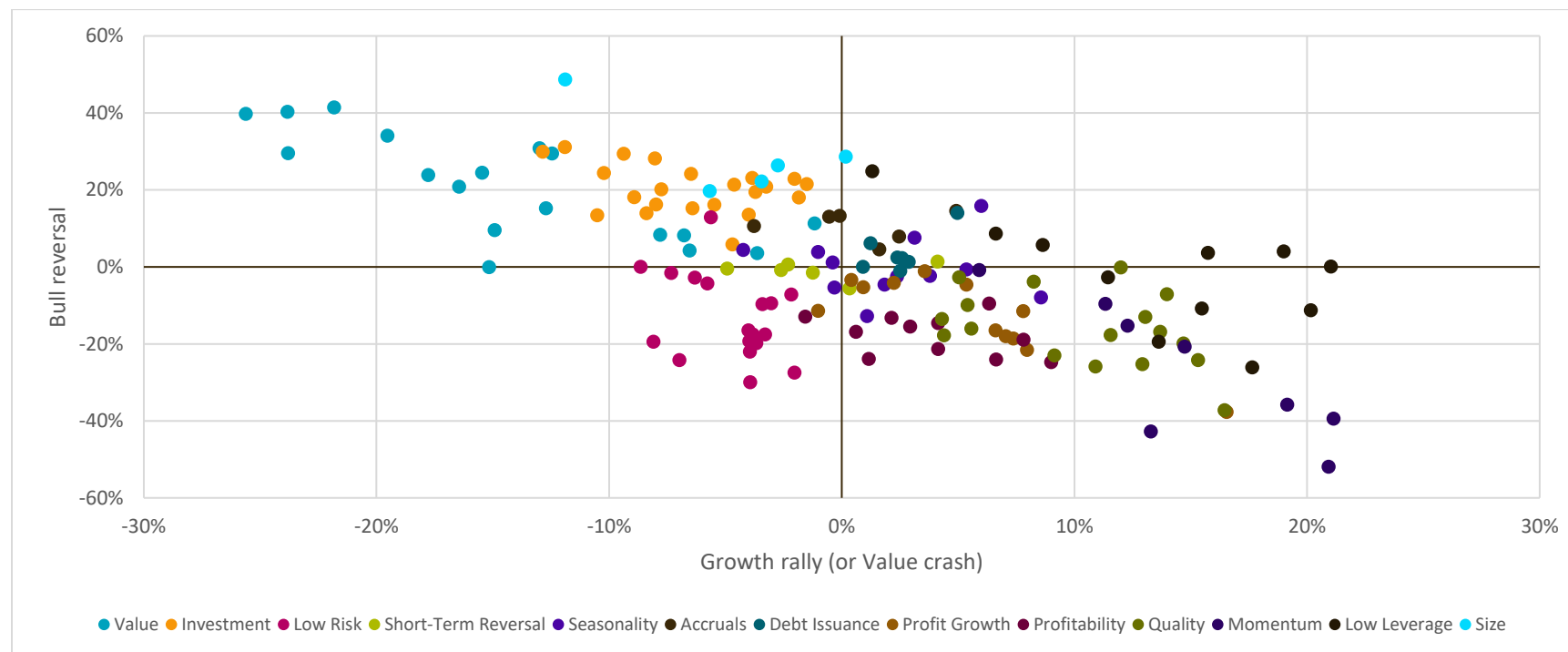


Exhibit 13: Factor sensitivity to the Quant Cycle: volatility of factor alphas in the different stages, 1963:07-2021:12

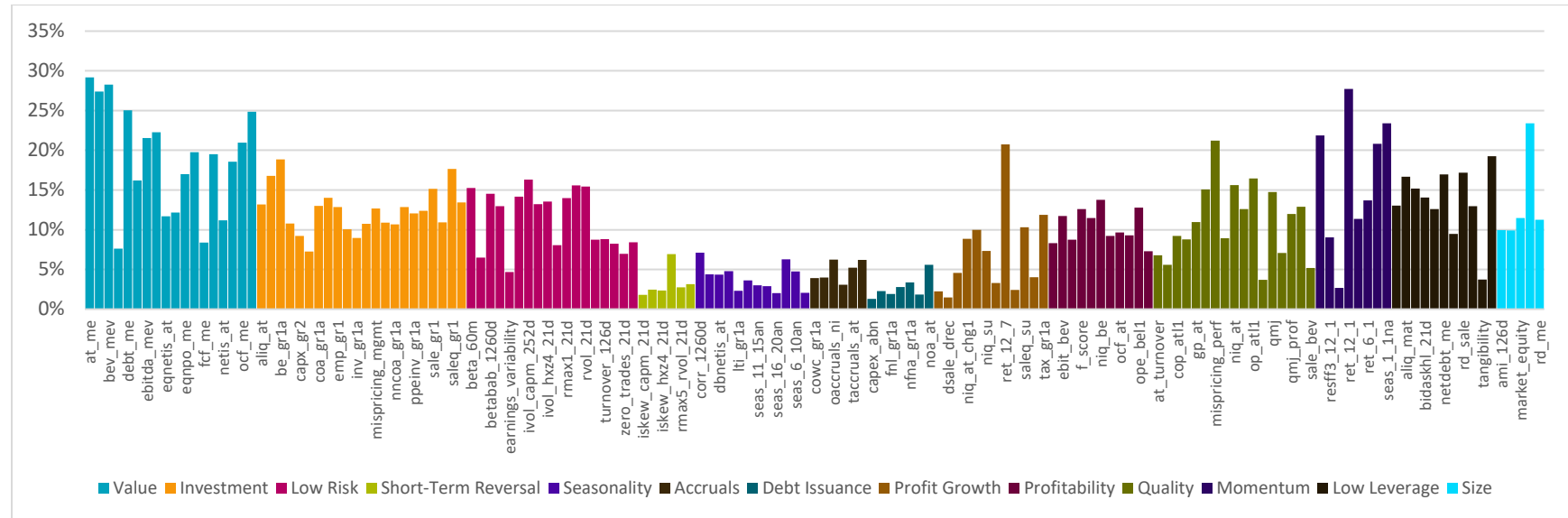


Exhibit 14: Performance of beta-sorted factor quintile portfolios, 1963:07-2021:12



Exhibit 15: Cumulative alpha of beta-sorted factor quintile portfolios, 1963:07-2021:12

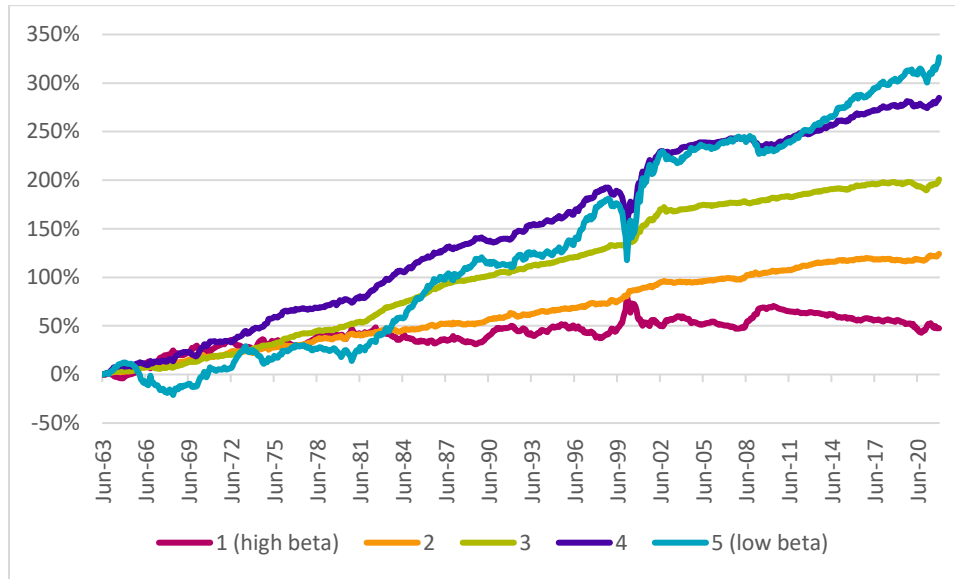


Exhibit 16: Cumulative return of seasonal momentum strategy applied to factors, 1963:07-2021:12

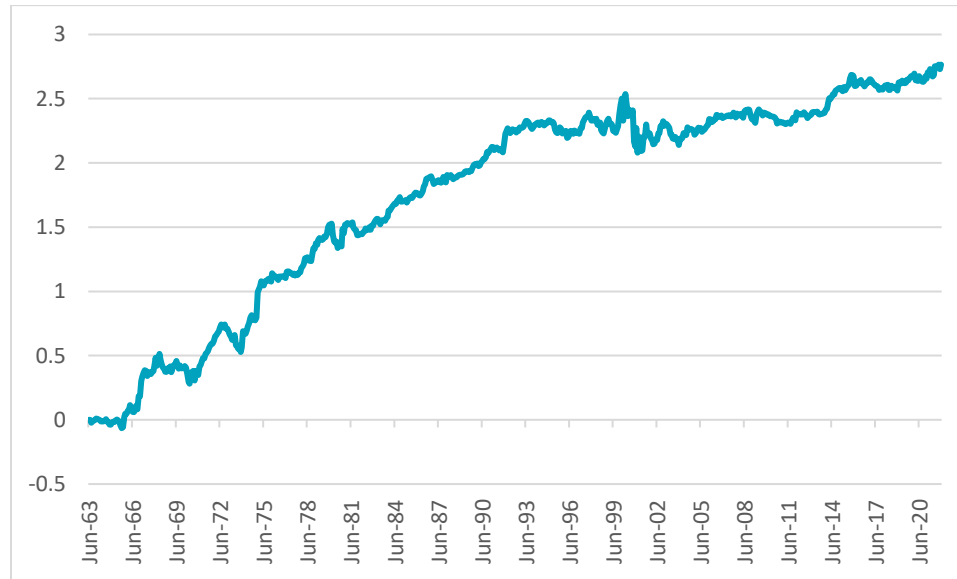


Exhibit 17: Autocorrelation in monthly factor returns, 1963:07-2021:12

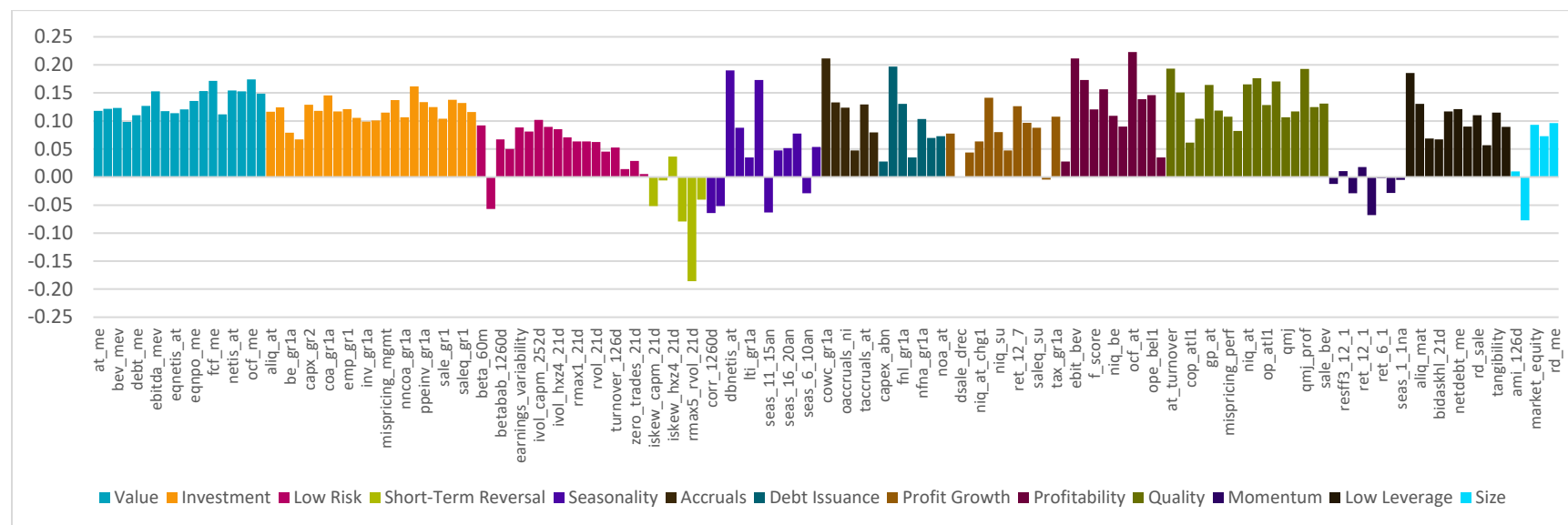


Exhibit 18: Cumulative return of 1-month momentum strategy applied to factors, 1963:07-2021:12

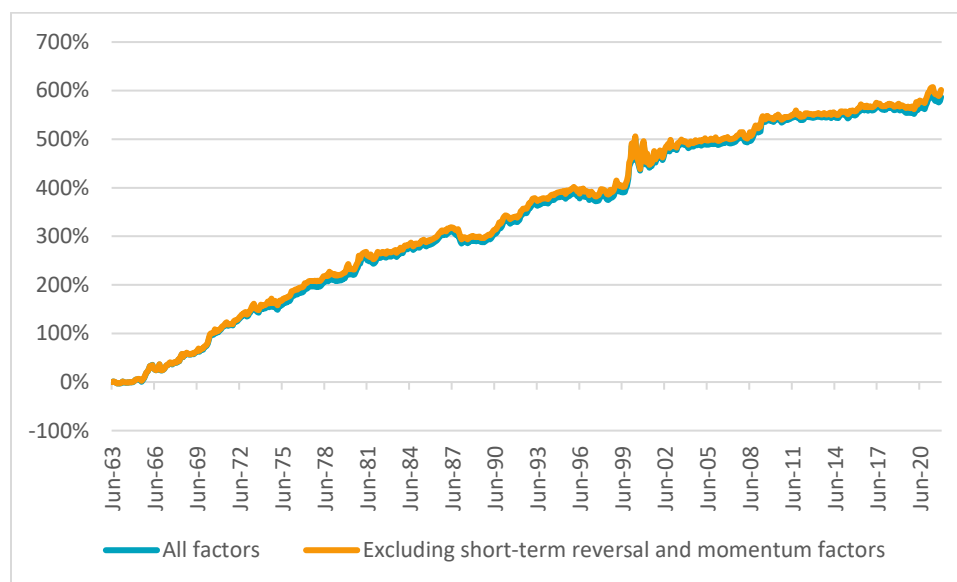


Exhibit 19: Alpha of factor rotation strategies over the Quant Cycle, 1963:07-2021:12

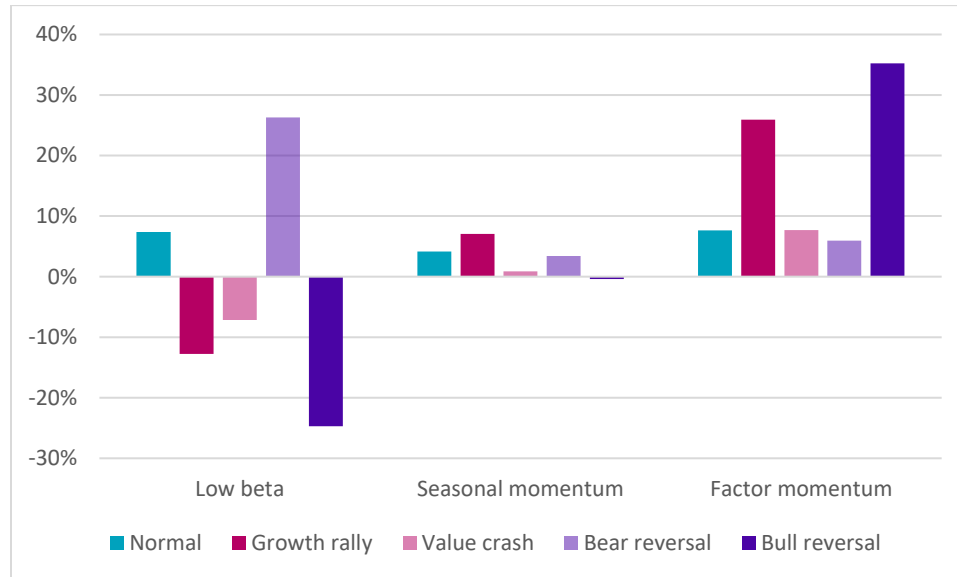


Exhibit 20: Qualitative summary

	Alpha	Beta	Lagged beta	January Effect	Quant cycle sensitivity	Turnover	Auto- correlation
Value	++	–	0	+	***	*	+
Investment	++	–	0	+	***	*	+
Low Risk	++	–	0	–	***	*	+
Short-Term Reversal	+	0	0	0	*	***	0
Seasonality	+	0	0	0	*	***	+
Accruals	+	–	0	0	*	*	+
Debt Issuance	+	0	0	0	*	*	+
Profit Growth	+	0	0	0	**	*	+
Profitability	+	–	0	–	***	*	+
Quality	+	0	0	0	***	*	+
Momentum	++	–	0	–	***	**	0
Low Leverage	0	+	0	0	***	*	+
Size	0	+	+	++	***	*	+